

Table 1 | Measurement Process^{96,105} cluster: definitions, examples and references

Dimension	Subdimension	Definition	Example	References
device error		The extent to which data values originating from a sensor are accurate and precise.		
	accuracy	The extent of closeness of data values to real values. (From Black and van Nederpelt ¹⁹⁷)	The distortion of the EEG signal due to patient movement.	57,75–78,80,82,84–91,93,94,96,98–109,112,113,128,129,132–134,148,151,152,160–168,172
	precision	The extent to which the error in data values spreads around zero (in statistics). (From Black and van Nederpelt ¹⁹⁷)	The difference between two haemoglobin level measurements from the same blood sample.	74,75,80,90,104–106,118,131,132,152,164,182
human-induced error		The extent to which manual inputs unintentionally influence the data in a wrong way.		88
	noisy labels	The extent to which labels provided by humans are accurate and precise.	The difference in the segmentation masks from two different radiologists due to different opinions or shaky hands.	84,101,102,106,118,120,131,132,143,146,154–159,169–176,178–183,185
	carelessness	The extent to which negligence or unintended human errors cause faulty data.	Persistent rounding of measurement readings: 120–80 systolic-diastolic heart pressure instead of actual 121–82 reading.	57,77,78,84,85,87,90–92,96,98–102,104,109,111,112,130,131,164,182
	outliers	The extent to which data values exist that do not match the expected distribution.	Age of a patient is greater than 150 years.	84,98,119,131
completeness		The extent to which data values are present. (From Black and van Nederpelt ¹⁹⁷)	Missing entry in dataset.	75–79,81–84,86–92,94–99,101–107,109,111,112,117,119,126–128,133,134,167,180,182
source credibility		The extent to which data values are regarded as true and believable in terms of their source. (Based on Black and van Nederpelt ¹⁹⁷)		57,75–78,80,83,101,104,106–110,113,120,183
	expertise	The extent of knowledge and experience with which the measurement was performed and the data was processed.	Diagnosis was done by a student from second semester vs. trained practitioner/physician.	85,90,98,104,106,112
	traceability	The extent to which data lineage is available. (From Black and van Nederpelt ¹⁹⁷)	The normalisation of columns is not documented or access to raw data is not provided.	57,74,75,80,83,105,106,110
	data poisoning	The extent to which data values are intentionally falsified.	The cohort is selected in a way that inflates the performance.	57,77,78,90,91,101,102,111,136,159,182

Table 2 | Timeliness cluster: definitions, examples and references

Dimension	Subdimension	Definition	Example	References
timeliness		The extent to which the ‘age’ and the ‘currency’ of data values are appropriate for the task at hand. (Based on Wang and Strong ⁵⁷)		79,84,88,90,105,109,113
	age	The extent to which the age of data values is appropriate for the task at hand. (Based on Wang and Strong ⁵⁷)	A 20-year-old dataset including measurements of patients cholesterol level is not a valid dataset for the task ‘predict the current risk of cardiovascular disease of the patient’.	57,75–77,80,82–85,89–91,100–103,106–108,120,133
	currency	The extent to which data values are up to date. (From Black and van Nederpelt ¹⁹⁷)	Updated medical knowledge leads to different diagnosis and thus label change.	75–78,80,86–88,90,93,96,99–101,103,105,106,108,109,111

necessarily better and adding noise to the training data might in some instances even improve performance^{200–202}. The errors belonging to the dimension *human-induced errors* are of a fundamentally different nature and need to be treated accordingly. This type of error includes human *carelessness* and *outliers* in the dataset due to (unintentional) human mistakes. The final subdimension, *noisy labels*, is one of the most relevant topics in current ML research^{157,159,169,182}. Since in the medical domain, supervised learning paradigms are prevalent, proper feature selection and reliable labelling are indispensable. However,

human decision making can be highly irrational and subjective, especially in the medical context^{203–205}, representing one of various sources of labelling noise²⁰⁶. Among expert annotators there is often considerable variability^{206,207}. Even in the most common (non-medical) datasets of ML (e.g., MNIST²⁰⁸, CIFAR-100²⁰⁹, Fashion-MNIST²¹⁰) there is a significant percentage of wrong labels^{211,212}. In contrast to the precision of instruments, noise in human judgements is demanding to be assessed through so-called noise audits to identify different factors, like pattern noise and occasion noise in the medical decision process²¹³.

Table 3 | Representativeness^{74,85,89,90,102–106,121} cluster: definitions, examples and references

Dimension	Subdimension	Definition	Example	References
variety		The extent to which the dataset is diverse.		57,113,120
	variety in demographics	The extent to which measured subjects are diverse.	Model is supposed to be used for people of all ages, but the dataset does not contain any patient under 30.	57,77,96,111,153,190
	variety of data sources	The extent to which data are available from different sources. (Based on Black and van Nederpelt ¹⁹⁷)	Data collected from one specific device but product is supposed to run on other devices, too (in terms of different built-in hardware).	111,128,137,143,188,191
depth of data		The extent to which the volume of the dataset is appropriate. (Based on Wang and Strong ⁵⁷)		57,77,78,84,104,190
	dataset size	The extent to which the quantity of records is appropriate.	A dataset of three observations is not a solid foundation for the classification of three different classes.	100,110,122–126,129,138,146,147,149,151–154, 156,158,159,169,183–189
	granularity	The extent to which the level of detail of the dataset and data values is appropriate.	The resolution of the CT image is too low to identify the tumour.	74,76,83,86,88,90,92,101,102,104,107,187,189
	coverage	The extent to which relevant subsets of the dataset satisfy the dimensions ‘variety’ and ‘target class balance’.	For the subset of a certain age class (e.g. 20–40 year olds) only healthy patient records exist although the disease exists in people of that age group in the real world.	57,74–77,85,91,92,96,100,102,104,105, 108,120,125,146
target class balance		The extent to which the classes of a target variable have similar size.	The dataset includes only one record of a patient with the rare disease that the model is supposed to predict.	115,116,122,138,140–142,144, 150,155,159

Table 4 | Informativeness⁸⁰ cluster: definitions, examples and references

Dimension	Subdimension	Definition	Example	References
understandability		The extent to which data values are clear without ambiguity and easily comprehended. (From Wang and Strong ⁵⁷)	Physician’s diagnosis (comment in natural language) is written ambiguously such that no explicit label/ICD code can be assigned.	57,74,76–78,80,83,85,88–90, 98,100,103,104,106,108,113
redundancy		The extent to which logically identical data are stored more than once. (From Black and van Nederpelt ¹⁹⁷)		84,103,110,150
	conciseness	The extent to which data values, attributes and records are compactly represented. (Based on Wang and Strong ⁵⁷)	One column for each possible diagnosis: Column ‘Normal’ (yes/no), ‘Myocardial Infarction’ (yes/no) instead of a single column ‘diagnosis’ (‘Normal’/‘Myocardial Infarction’).	57,76–78,80,83,84,91,95,104,106,111
	uniqueness	The extent to which records occur only once in a data file. (From Black and van Nederpelt ¹⁹⁷)	Same record occurs more than once.	74–76,76,77,82,84,93–95,97,98,100,102, 103,105,106,111,114,116,118–120,128,155
informative missingness		The extent to which missing data values provide useful information.	If ‘cholesterol level’ is empty in EHR patient record, then it is likely, that the doctor never felt the need to measure it. This could indicate, that the risk for a cardiovascular disease is low.	96,98,99,105,111,119,126,135,182
feature importance		The extent to which the attributes are beneficial and provide advantages from their use. (Based on ‘usefulness’ from Wang and Strong ⁵⁷)	The feature ‘insurance company’ carries less information relevant for the prediction of a stroke than the feature ‘blood pressure’.	57,74,76–78,80,85,88,90,91,96,100–109, 111,112,120,143,149,150,153,187

Such intra- and inter-observer variability has always been a highly important topic in many medical disciplines, e.g., in radiology where guidelines, training and consensus reading approaches are used to reduce noise²¹⁴.

Another issue that frequently occurs in the data acquisition process and which plays an important role in ML is the absence of data values with unknown reason. We follow the ML vocabulary by capturing this quality issue with the dimension *completeness*, while noting

that outside of ML contexts, this term is commonly used to describe representativeness, coverage or variety in other contexts. Most prominently, Wang et al.⁵⁷ define completeness as ‘breadth, depth, and scope of information’. This definition has been picked up by other researchers, as well^{100,106,126}. In ML, however, *completeness* is usually measured by the ratio of missing to total values. Apart from the mostly quantitative dimensions within the cluster, the dimension *source credibility* is concerned with mostly qualitative characteristics. On the

Table 5 | Consistency^{76,82–84,89,90,98,101,103–106,120} cluster: definitions, examples and references

Dimension	Subdimension	Definition	Example	References
rule-based consistency		The extent to which data values comply with a rule. ('Validity' from Black and van Nederpelt ¹⁹⁷)		
	syntactic consistency	The extent to which data values comply to a dictionary and are always presented in the same format. (Based on Wang and Strong ⁵⁷)	A mixture of 'm' and 'male' violates syntactic consistency.	57,74,75,77–84,90–98,100–106,108,109,111–113,116,119,120,127,133
	compliance	The extent to which the dataset and data values are in accordance with laws, regulations or standards. (From Black and van Nederpelt ¹⁹⁷)	All entries of the dataset follow the ICD-10-CM coding.	75,80,81,83,85,90,96,98,102,107,111
logical consistency		The extent to which data values are plausible and records are semantically consistent.		106
	semantic consistency	The extent to which records are free of contradictions.	'Abnormal radiologic findings on diagnostic imaging of left kidney', but the left kidney was donated before this diagnosis was made.	74,75,77,80,81,83–87,90–96,98,99,101–108,110–113,117,120,128
	plausibility	The extent to which data values match knowledge of the real world. (From Black and van Nederpelt ¹⁹⁷)	Patient is 5 years old and 180cm tall.	57,75,77,78,87,93,95,97,99,102,103,105,106,112,120,128
distribution consistency		The extent to which distributions are stable.		139,177
	homogeneity	The extent to which distributions are stable among relevant subsets.	Data originates from two devices with different underlying calibrations.	74,84,102,144,171,186
	distribution drift	The extent to which new data matches the distribution of existing data.	The median age of the population shifts over the decades.	75,77,79,85,93,100,102,145,192

Table 6 | Data Management cluster: definitions and references

Dimension	Definition	References
documentation	Metadata about the dataset.	74,75,77,85,89,96,102,104,106–108,111
security	A characteristic that ensures that unauthorised changes to the dataset are prohibited.	57,77,78,80,89,90,103,104,106,109,113
privacy	A characteristic that ensures that the dataset is recorded and stored in a way that protects personal data.	75,83,89,90,104,109–111,165,167
FAIR principles	The FAIR Principles ¹⁹⁹ require a dataset to be findable, accessible, interoperable and reusable.	57,74–78,80,83,85,88,90,100–104,106,108–113

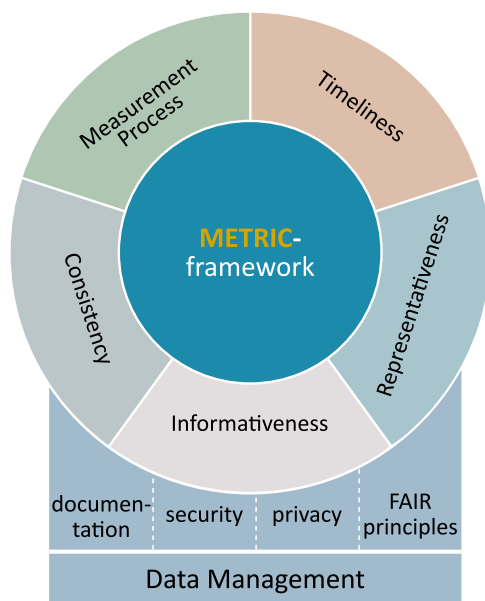


Fig. 4 | METRIC-Framework in relation to data management. The cluster *data management* is concerned with the effective usage of the dataset. It includes basic requirements for the dataset but does not address data quality issues regarding its content. Therefore, it can be seen as a prerequisite for assessment using the METRIC-framework. Figuratively speaking, the *data management* cluster serves as a stable foundation for the wheel of data quality.

one hand, it includes the question whether or not the measured data can be trusted based on the *expertise* of people involved in data measurement, processing and handling. On the other hand, the subdimension *traceability* evaluates whether changes from original data to its current state are documented. Being aware of modifications such as the exclusion of outliers, automated image processing in medical imaging or data normalisation and their utilised algorithms are necessary for understanding the composition of the data. Finally, the subdimension *data poisoning* considers whether the data was intentionally corrupted (e.g., adversarial attacks) to cause distorted outcomes. The entire cluster *measurement process* is crucial for data quality evaluation in the medical field since errors may propagate through the ML model and lead to false diagnosis or treatment of patients.

We note that special consideration has to be given to the field of medical imaging within the *measurement process* cluster due to the fact that many imaging devices are not classical measurement devices. For instance, in current radiological practice, decisions are still based mainly on visual inspection of images and rating of diseases and therapy effects are often done in qualitative terms such as 'enlarged', 'smaller' or 'enhanced'. This places a lot of importance on the qualitative subdimensions *source credibility* and *expertise*, with respect to quality assessment in such use cases. However, over the last two decades significant efforts have been made to establish quantitative imaging biomarkers to transform scanners more into measurement devices to quantify biophysical parameters, like flow, perfusion, diffusion or elasticity. Such quantitative imaging approaches reduce the operator dependency and enable more quantitative evaluation in the dimension *device error*. Worldwide alliances such as Quantitative